Unix for Developers: Trial Exam

Part 1: Multiple Choice (Choose the correct answer)

1. **What does the `ls` command do in Unix?**

    a) Lists the contents of a file
    b) Lists the contents of a directory
    c) Creates a new directory
    d) Removes a file

2. **Which command is used to change the current working directory in Unix?**

    a) `chdir`
    b) `cd`
    c) `pwd`
    d) `mv`

3. **Which of the following commands will list all files, including hidden files, in a directory?**

    a) `ls -a`
    b) `ls -l`
    c) `ls -h`
    d) `ls -r`

4. **What does the `chmod` command do?**

    a) Changes file ownership
    b) Changes file permissions
    c) Creates a symbolic link
    d) Changes file content

5. **Which command would you use to display the current working directory?**

    a) `cd`
    b) `pwd`
    c) `ls`
    d) `locate`

6. **What is the purpose of the `grep` command in Unix?**

    a) To search for a specific string within a file
    b) To display a file's contents

c) To remove files
d) To count lines in a file

7. **Which of the following commands will display the disk usage of files and directories?**

   a) `df`
   b) `du`
   c) `ls -l`
   d) `stat`

8. **What does the `ps` command do in Unix?**

   a) Displays the process status
   b) Prints to the screen
   c) Lists directories
   d) Displays system uptime

Part 2: True or False

9. **The `echo` command is used to output a string to the terminal.**

   - True
   - False

10. **In Unix, files are not case-sensitive.**

    - True
    - False

11. **The `cat` command can only be used to display the contents of a file and cannot concatenate files.**

    - True
    - False

12. **The `cp` command can be used to move files between directories.**

    - True
    - False

13. **The `find` command is used to search for files based on specific criteria in the filesystem.**

- True
- False

14. **The `head` command displays the first 10 lines of a file.**
    - True
    - False
15. **In Unix, directories can be removed with the `rm` command if they are not empty.**
    - True
    - False

Part 3: Short Answer

- **Explain the difference between a relative and an absolute file path.**
- **What is the purpose of the `shebang` line in a shell script?**
- **Explain the difference between the `rm` and `rmdir` commands.**
- **What is the `|` (pipe) operator used for in Unix? Provide an example.**
- **What command would you use to find all the files in your current directory that are larger than 1MB?**

Part 4: Practical Tasks

Task 1: File and Directory Operations

1. Create a new directory named `exam_files` in your home directory.
2. Within `exam_files`, create a file called `test_file.txt` and add the text `Hello, World!` to it.
3. Move `test_file.txt` to a new directory called `archive` inside `exam_files`.
4. Copy `test_file.txt` back into the `exam_files` directory and rename it to `test_file_copy.txt`.
5. Delete `test_file.txt`.

Task 2: Shell Scripting

Create a shell script that:

1. Prints the message “Welcome to the trial exam!”.
2. Accepts one argument (a file name) and checks if the file exists.
    - If the file exists, print `File [filename] exists.`
    - If the file does not exist, print `File [filename] does not exist.`
3. The script should be executable.

Task 3: Using Redirection and Pipes

1. Create a file named `file1.txt` and add some text to it using the `echo` command. For example, `echo "This is file 1"> file1.txt`.
2. Create another file `file2.txt` with similar content.
3. Use the `cat` command to display the contents of both files by redirecting the output to a third file called `merged.txt`.
4. Use the `grep` command to search for the word “file” in the `merged.txt` file and display the result.
5. Redirect the output of the `grep` command into a new file `result.txt`.

Part 5: Bonus (Advanced) Questions

1. Write a shell script that counts the number of lines in all `.txt` files in the current directory and outputs the total count.
2. Using the `find` command, list all files in the `/home` directory that were modified in the last 7 days.

Part 6: Syntax Errors in Shell Scripts

The following shell script contains **3** syntax errors. Find the errors by providing the line they appear in. Write the correct line next to it.

```
#!/bin/sh
$counter=1
number1=1
number2=$number1
echo $number1
echo $number2
while [ counter -lt 20 ]; do
    result=`expr $number1 + $number2`
    echo $result
    number1=$number2
number2=$result
counter=`expr $counter++`
done
```

What does the script do when all the errors are fixed?